



Systems Laboratory

Information and Digital Inclusion

LUDOLAB

DIGITAL INCLUSION PROJECT

MODALITY
VIRTUAL AND IN-Person

USER MANUAL

DIGITAL SIGNATURE

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1. Project Objectives

Train teachers from the Ministry of Education in the use of ICT tools to improve their teaching performance, in order to promote educational innovation and online learning.

Train priority attention groups or digitally illiterate groups in the use of ICT tools to apply what they have learned in their daily lives, promoting their digital inclusion and improving their quality of life.

2. Objective of the Manual

Provide a complete and detailed guide on the use of digital signature, including step-by-step instructions for its implementation and management in various types of electronic documents.

Explain clearly and concisely the benefits and security aspects of the digital signature, helping users understand how it protects the integrity and authenticity of documents.

3. Scope

This document covers all stages of using the digital signature, from its definition and basic concepts to configuration, practical use, resolution of common problems and good practices. It is designed to be a complete reference for any user who needs to use digital signature in their work or personal environment.

4. Addressed to

This manual is aimed at all people, especially those who have low technological knowledge or are technologically illiterate. Its purpose is to provide a simple and accessible guide so that any user, regardless of their previous experience with the technology, can understand and use the digital signature effectively.

5. Application introduction

The digital signature application allows users to sign electronic documents in a secure and legally valid manner. Its main functions include creating digital signatures, verifying the authenticity of signed documents, and managing digital certificates. The application is designed to be intuitive and easy to use, providing a reliable solution to ensure the integrity and authenticity of documents in digital format.

6. Basic notions

Before starting with the digital signature, it is important to understand some basic concepts and elements:

Application workspace: When opening the application, users will see an interface divided into several sections:

Document panel: Here the documents you want to sign are loaded and displayed.

Signature panel: Shows the available digital signatures and allows them to be managed.

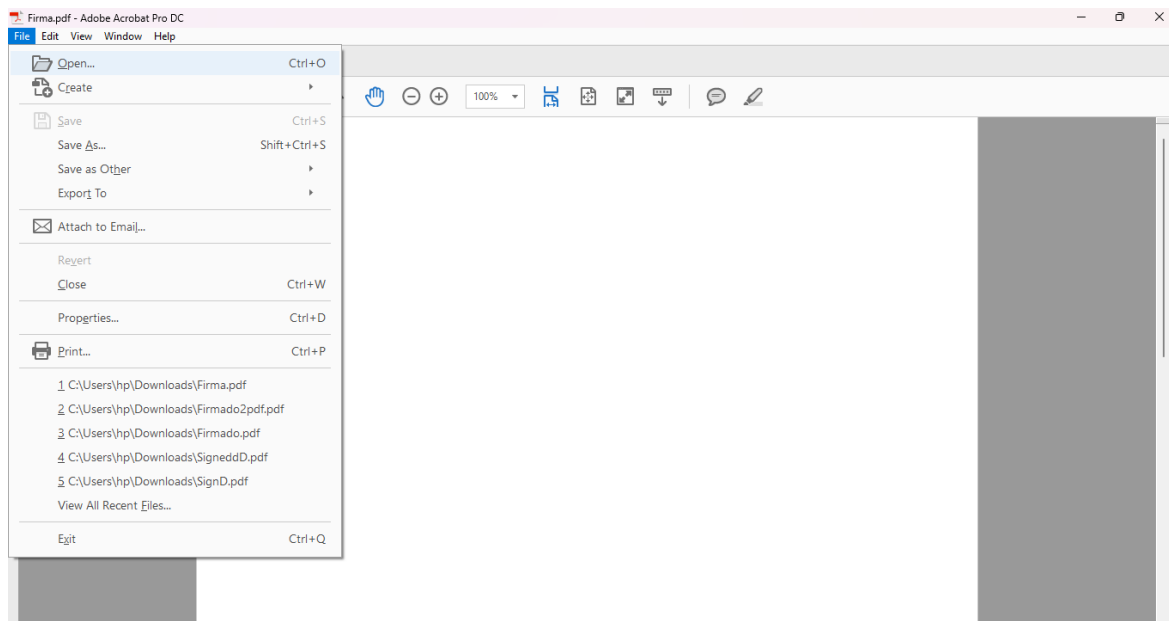
Settings panel: Allows you to adjust application options, such as digital certificate selection and security preferences.

Digital Certificates: A digital certificate is a file that is used to create a digital signature. This certificate contains user information and a public key, which is essential for signature verification.

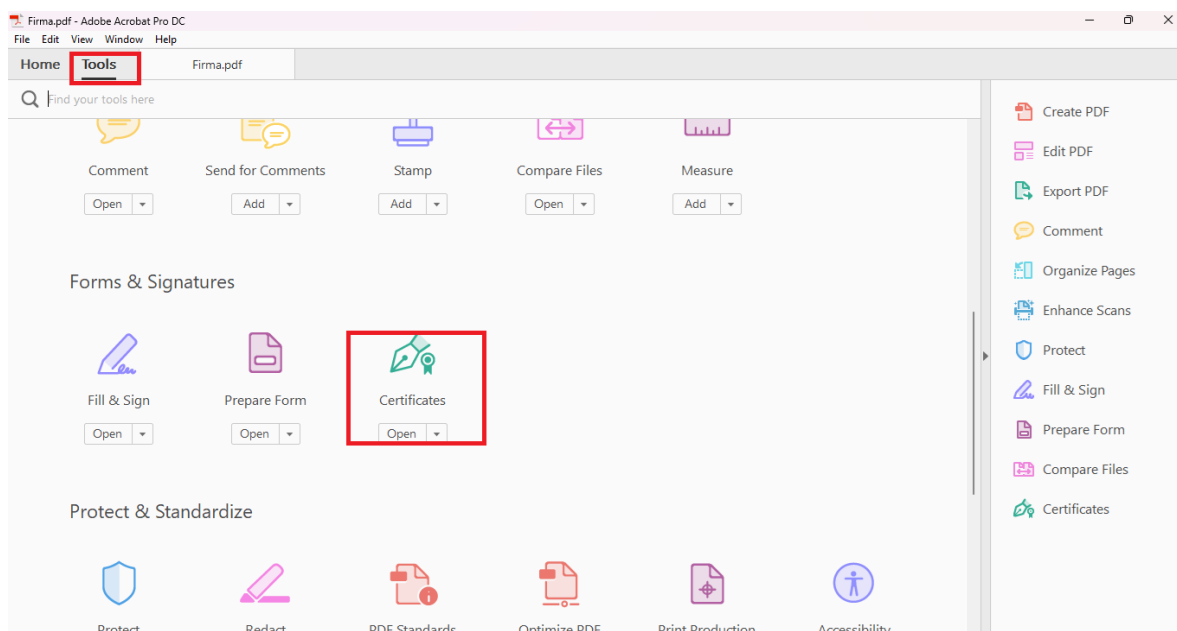
Digital signature process: To sign a document, the user must select the document, apply their digital signature using the corresponding certificate and finally save the signed document. The digital signature ensures that the document has not been altered from the moment of signing.

7. Steps to follow

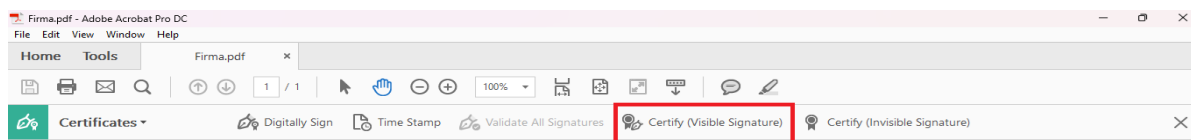
The first thing to do is open the document to be signed in Adobe Acrobat.



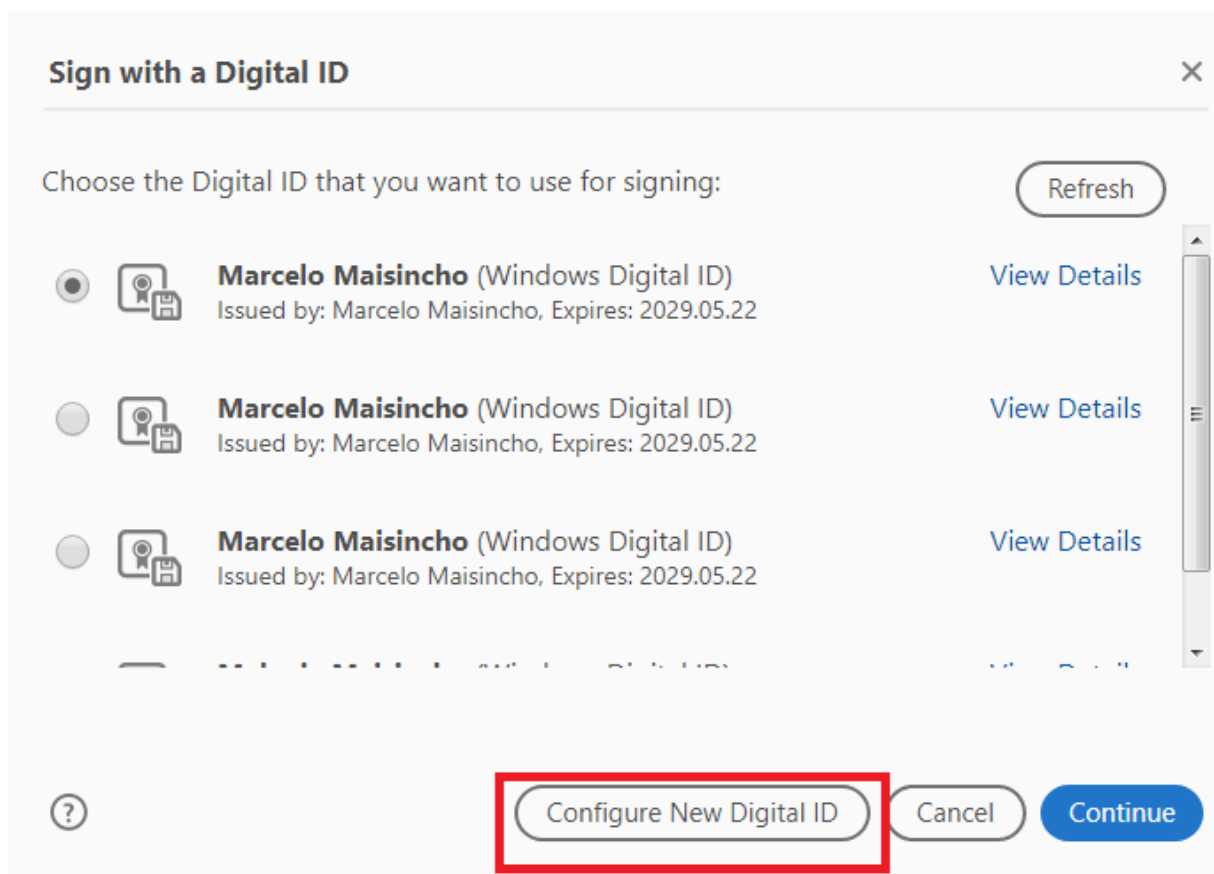
Subsequently, click on the Tools option and select the certificates option.



Then select the certify option and draw a rectangle in the area you want to sign.



Select the indicated option.



Choose the option to create a new digital ID.

Configure a Digital ID for signing

A Digital ID is required to create a digital signature. The most secure Digital IDs are issued by trusted Certificate authorities and are based on secure devices like smart card or token. Some are based on files.

You can also create a new Digital ID, but they provide a low level of identity assurance.

Select the type of Digital ID:

- ☐ **Use a Signature Creation Device**
Configure a smart card or token connected to your computer
- ☐ **Use a Digital ID from a file**
Import an existing Digital ID that you have obtained as a file
- ☒ **Create a new Digital ID**
Create your self-signed Digital ID

 Cancel Continue

Click on the save in Windows option.

Select the destination of the new Digital ID

Digital IDs are typically issued by trusted providers that assure the validity of the identity. Self-signed Digital ID may not provide the same level of assurance and may not be accepted in some use cases.

Consult with your recipients if this is an acceptable form of authentication.

☐ **Save to File**
Save the Digital ID to a file in your computer

☒ **Save to Windows Certificate Store**
Save the Digital ID to Windows Certificate Store to be shared with other applications

 Back Continue

Fill out the fields with your information and save.

Create a self-signed Digital ID

Enter the identity information to be used for creating the self-signed Digital ID.

Digital IDs that are self-signed by individuals do not provide the assurance that the identity information is valid. For this reason they may not be accepted in some use cases.

Name	Enter Name...
Organizational Unit	Enter Organizational Unit...
Organization Name	Enter Organization Name...
Email Address	Enter Email...
Country/Region	US - UNITED STATES
Key Algorithm	2048-bit RSA
Use Digital ID for	Digital Signatures

?
Back
Save

Click continue

Sign with a Digital ID

Choose the Digital ID that you want to use for signing:

☒

Cristian Rosero (Windows Digital ID)
Issued by: Cristian Rosero, Expires: 2029.05.22

[View Details](#)

☐

Marcelo Maisincho (Windows Digital ID)
Issued by: Marcelo Maisincho, Expires: 2029.05.22

[View Details](#)

☐

Marcelo Maisincho (Windows Digital ID)
Issued by: Marcelo Maisincho, Expires: 2029.05.22

[View Details](#)

?
Configure New Digital ID
Cancel
Continue

Select the sign option.

Sign as "Cristian Rosero"

Appearance

Created 2024.05.21 23:41:37 -05'... ▾

Create

Edit

Cristian
Rosero

Certified by
Cristian Rosero
Date: 2024.05.22
01:29:12 -05'00'

[View Certificate Details](#)

Reason

none ▾

Location

Contact Info

Permitted Actions After Certifying

Form fill-in and digital signatures ▾

Review document content that may affect signing

Review

Back

Sign

Signature made.

Cristian
Rosero

Digitally signed
by Cristian
Rosero
Date: 2024.05.22
01:29:57 -05'00'

8. Frequent questions

What is a digital signature and how does it work?

A digital signature is a cryptographic tool that ensures the authenticity and integrity of an electronic document. It works using a digital certificate that contains the user's public key. When a document is digitally signed, a hash (cryptographic digest) of the document is created, which is encrypted with the signer's private key. The recipient of the document can verify the authenticity of the signature by decrypting the hash with the signer's public key and comparing it to a generated hash of the received document.

Is a digital signature legally valid?

Yes, a digital signature is legally valid in many countries, as long as it complies with the specific regulations and regulations of each jurisdiction. In general, a digital signature must be backed by a certificate issued by a recognized certification authority and meet established standards to be considered legally binding.

How can I obtain a digital certificate?

A digital certificate can be obtained through a recognized Certification Authority (CA). The process usually involves verifying the applicant's identity through identification documents and sometimes in-person or online validation. Once the identity is verified, the CA issues a digital certificate that the user can install on their computer or mobile device.

What should I do if I lose my private key?

If you lose your private key, you will not be able to sign documents or decrypt messages encrypted with that key. It is crucial to keep a backup of your private key in a safe place. If the private key is permanently lost, you will need to obtain a new digital certificate from your Certification Authority and revoke the old certificate to prevent misuse of the key.

How do I verify the authenticity of a digital signature on a received document?

To verify the authenticity of a digital signature, the signer's digital certificate is used. Most digital signature applications and electronic document management programs have built-in features to verify digital signatures. When opening a digitally signed document, the program will verify the signature using the signer's public key, ensuring that the document has not been altered and that the signature is valid.

9. Glossary

Término	Descripción
Certification Authority (CA)	Entity responsible for issuing and managing digital certificates. A CA verifies the identity of people or entities before issuing them a digital certificate.
Digital certificate	Electronic document issued by a Certification Authority that links a public key with the identity of the owner. It is used to verify the authenticity of a digital signature.
Private Key	Component of a cryptographic key pair that must be kept secret. It is used to create digital signatures and decrypt encrypted data.
Electronic Document	Document in digital format that may contain text, images, data, etc. Unlike paper documents, electronic documents can be digitally signed to ensure their authenticity and integrity.
Digital signature	Cryptographic tool that ensures the authenticity and integrity of an electronic document. It uses a digital certificate to create an encrypted hash of the document.
PDF	File format designed to present documents consistently across different devices and platforms. PDF documents may contain digital signatures.
Security of the information	Practices and technologies used to protect information from unauthorized access, alteration, disclosure or destruction.
USB Token	Physical device that stores cryptographic keys and digital certificates, providing a secure way to use digital signatures.

10. Bibliography and References

Reference	Title
Adobe Systems Incorporated (2023)	Guide to Digital Signatures
U.S. National Institute of Standards and Technology (NIST) (2020)	Digital Signature Standard (DSS)
GlobalSign (2021)	Digital Certificates and PKI Explained